DHANANJAI SHARMA

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SUMMARY

Artificial Intelligence and Cybersecurity Research Engineer specializing in AI security, adversarial machine learning, trustworthy AI, and privacy-preserving AI. Experienced in end-to-end research spanning threat modeling, method development, implementation, and experimental evaluation, with current work on Vision-Language-Action security and privacy-preserving LLM/AI-agent inference using FHE. Proficient in Python, PyTorch, LLMs, diffusion models, NLP, and generative AI, with experience translating research ideas into reproducible prototypes and technical manuscripts.

CORE COMPETENCIES

- **Programming:** Python, C, C++, Java
- **ML Frameworks & Tools:** PyTorch, Hugging Face Transformers, CUDA, Jupyter, Git
- **Machine Learning:** Deep Learning, Neural Networks, Graph Machine Learning, Reinforcement Learning
- **Generative AI:** LLMs, RAG, LoRA/PEFT, Diffusion Models, GANs, VAEs
- **AI Security:** Adversarial ML, Trustworthy AI, LLM Security, VLA/Embodied AI Security, Jailbreak Evaluation
- **Privacy-Preserving AI:** Fully Homomorphic Encryption (FHE), Privacy-Preserving LLM Inference, Secure AI Agents
- **Cybersecurity:** VAPT, Network Security, Information Security, Threat Modeling, Security Evaluation, Digital Forensics
- **NLP & Information Extraction:** NLP, OCR, Document Understanding, Information Extraction, Domain-Specific LLMs
- **Research:** Literature Review, Experimental Design, Ablation Studies, Benchmarking, Technical Writing

WORK EXPERIENCE

Cybersecurity Research Engineer **Dec 2025 - Present**

ST Engineering (Singapore)

- Identify research gaps in AI security and privacy, formulate threat models and research hypotheses, and develop technical approaches for evaluating and mitigating emerging AI-system vulnerabilities.
- Research the adversarial robustness of Vision-Language-Action (VLA) models and embodied AI systems, focusing on visual and textual attacks and inference-time detection and mitigation techniques.
- Design and implement Python/PyTorch experimental pipelines for adversarial evaluation, robustness benchmarking, ablation studies, and quantitative analysis across attack and defense configurations.
- Investigate privacy-preserving LLM and AI-agent inference using Fully Homomorphic Encryption (FHE), developing methods to protect prompts, intermediate representations, tool interactions, and agent communication while reducing cryptographic overhead and preserving model utility.
- Collaborate with cross-functional research and engineering teams to conduct technical surveys, develop research prototypes, analyze experimental findings, and prepare manuscripts and technical reports for peer-reviewed security and AI venues.

AI Engineer Intern **May 2025 - Aug 2025**

Integro Technologies Pte Ltd (Singapore)

- Built a domain-specific LLM/RAG system for corporate lending, achieving >95% answer accuracy across ~1500 financial documents and reducing multi-query analysis time by ~70%.
- Developed OCR and PDF-to-Markdown pipelines for 10+ document types, including handwritten and scanned content, achieving >90% extraction accuracy and ~98% structure preservation.
- Helped deploy the organization's first AI-enabled financial analysis workflow, implementing role-based access control, encryption, protected document retrieval, and secure handling of model inputs and outputs for sensitive financial data.

Artificial Intelligence Research Intern **Jul 2023 - Jul 2024**
Armur.AI
- Led development of various AI models for use cases such as detecting and resolving vulnerabilities in blockchain smart contracts, automated penetration testing, and network security
- Delivered production-ready models with improved detection accuracy and efficiency, contributing to automation of critical cybersecurity workflows

EDUCATION

National University of Singapore **Aug 2024 – Jan 2026**
Master of Computing in Artificial Intelligence
- Courses: Neural Network and Deep learning; Graph machine Learning; Uncertainty Modelling in AI; Social Media Computing, Trustworthy Machine Learning, AI Planning and Decision Making
- Dissertation: *Revealing Learned Concepts in LoRA Text-to-Image Models*

SRM Institute of Science & Technology **Jun 2020 - Jul 2024**
Bachelor of Technology in Computer Science (Cyber security) (GPA: 9.43/10)
- Courses: Computer Programming ; Data Structures and Algorithms ; Operating Systems ; Networking ; Cyber Security Basics ; Information Security ; Artificial Intelligence
- Thesis: *Ethical Implications of AI in Cybersecurity: Machine Learning-Based Firewalls*

ACADEMIC PROJECTS

Diffusion-Based Adaptive Game Level Generation
- Designed and implemented difficulty-conditioned generative models for procedural game-level generation, comparing Stable Diffusion fine-tuning, LoRA, scratch-trained diffusion, and conditional VAEs on 25,000 procedurally generated mazes.
- Formulated a solver-based difficulty metric using shortest-path length and search complexity to guide maze generation.

Defending Large Language Models Against Jailbreak Exploits
- Developed and evaluated a multi-layer LLM security framework combining prompt sanitization, inference-time logit steering, and multi-agent defenses against jailbreak attacks.
- Evaluated aligned and unaligned Llama-based models on XSTest and in-the-wild jailbreak benchmarks, reducing attack success rate by *43%* using activation steering and achieving *0%* ASR with a domain-specific multi-agent defense.

PUBLICATIONS
- Spectral Detection and Defense against Adversarial Attacks in Vision-Language-Action Policies, *Under review as a journal paper at IEEE Transactions on Dependable and Secure Computing*
- Undisclosed Concept Discovery in LoRA Adapters for Diffusion-Based AIGC Models , *Under review as a conference paper at USENIX Security 2027*
- Automated Vulnerability Detection in Solana Smart Contracts Using a Fine-Tuned Language Model, JETIR, Vol. 11, Issue 4, April 2024
- Automated Penetration Testing using Large Language Models, IJSR, Vol. 13, Issue 4, April 2024
- Advancing Cybersecurity Through AI: A Multilayered Phishing Detection Approach, IJCRT, Vol. 11, Issue 12, December 2023

ADDITIONAL INFORMATION
- Achievements: Ranked in the top 10% of Bachelor of Technology cohort
- Tools & Frameworks: Kali Linux, React, Jupyter, LaTeX, Git, OpenSSH, AWS, SQL
- Languages: Fluent in English, Hindi & Japanese (basic; JLPT N5)
- Interests: Competitive Programming, Basketball